

HIGH-INTENSITY INTERVAL TRAINING EVERY SECOND WEEK MAINTAINS $\dot{V}O_{2\max}$ IN SOCCER PLAYERS DURING OFF-SEASON

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ABSTRACT

Slettaløkken, G and Rønnestad, BR. High-intensity interval training every second week maintains $\dot{V}O_{2\max}$ in soccer players during off-season. *J Strength Cond Res* 28(7): 1946–1951, 2014—Reduced endurance training among semiprofessional soccer players during off-season may have negative effect on game performance during the competition season. This negative effect can be prevented by adding high-intensity interval training (HIT) to normal activity. In this study, we want to compare 2 different frequencies of HIT (5 bouts of 4 minutes on 87–97% peak heart rate) session on maintenance of aerobic fitness among semiprofessional soccer players during a 6-week off-season period. Seventeen male players at second and third highest soccer division in Norway participated. The subjects were randomized into 1 HIT session every second week (HIT 0.5) or 1 HIT session per week (HIT 1). All participants performed a 20-m shuttle run test and a maximal oxygen uptake ($\dot{V}O_{2\max}$) test on treadmill before and after the training intervention. $\dot{V}O_{2\max}$ (HIT 0.5, 63.4 ± 5.9 ml·kg⁻¹·min⁻¹; HIT 1, 65.6 ± 2.1 ml·kg⁻¹·min⁻¹) and 20-m shuttle run performance (HIT 0.5, 2335 ± 390 m, HIT 1, 2531 ± 106 m) were not different between the groups before the training intervention. $\dot{V}O_{2\max}$ was maintained after the training intervention in both HIT 0.5 and HIT 1 (64.0 ± 5.9 ml·kg⁻¹·min⁻¹ and 64.3 ± 1.3 ml·kg⁻¹·min⁻¹, respectively). There was a reduction in distance covered during the 20-m shuttle run test in HIT 1 and when groups were pooled ($-7.9 \pm 5.7\%$ and $-6.4 \pm 7.9\%$, respectively, $p \leq 0.05$). In conclusion, HIT 1 did not maintain $\dot{V}O_{2\max}$ better than HIT 0.5 when added to normal off-season activity. However, performance in 20-m shuttle run, which is a more soccer-specific fitness test than $\dot{V}O_{2\max}$ test, was slightly reduced when both groups was pooled.

KEY WORDS aerobic fitness, beep-test, endurance training

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INTRODUCTION

One determining factor for aerobic fitness is maximal oxygen uptake ($\dot{V}O_{2\max}$), and it has been observed that increased $\dot{V}O_{2\max}$ also increases time spent on high-intensity running during a soccer match (8). It therefore seems clear that sufficient training for aerobic fitness is important for performance in soccer. For soccer teams, not participating in the highest divisions, off-season can last for about 4–7 weeks, and the training during this period is entrusted to the players themselves. Deconditioning during the off-season period has previously been reported (3) possibly caused by reduced training volume. If the players enter the preseason in a deconditioned state, they have to put lot of efforts in reaching optimal in-season aerobic fitness level. This may, on one hand, lead to less time available for soccer-specific technical and tactical training during the preseason and on the other hand, increase the risk of overtraining syndrome towards the end of the season (13). Maintaining aerobic fitness level during the off-season is therefore important, but the time available for such training can be scarce because of job and family situation. For lower division soccer players, knowing how much training that is needed to sustain $\dot{V}O_{2\max}$ through off-season is of great value. An early study on preservation of $\dot{V}O_{2\max}$ with reduced training frequency found that an initial increase in $\dot{V}O_{2\max}$ was maintained when frequency of sustained aerobic training was reduced from 4 to 2 days per week (10). High-intensity interval training (HIT) is however, more potent in increasing $\dot{V}O_{2\max}$ than lower intensity-sustained training (9,26). It is therefore possible that when using HIT, the frequency needed to maintain aerobic fitness could be even lower than 2 days per week. Knowing the HIT frequency necessary to avoid deconditioning during off-season in semiprofessional soccer players could be of great value for the players and for the coaches preparing for the next season.

This study aims at finding the HIT frequency that is needed for semiprofessional soccer players to maintain aerobic fitness during off-season. Two different frequencies of HIT were compared. We hypothesized that adding HIT once every week to normal off-season activity is sufficient to maintain aerobic fitness, as evaluated by $\dot{V}O_{2\max}$ and

performance on 20-m shuttle run test. We further hypothesized that adding HIT only once every second week to normal off-season activity is not sufficient to maintain aerobic fitness.

METHODS

Experimental Approach to the Problem

To find the HIT frequency necessary to avoid deconditioning, players from the second and third highest soccer division in Norway were trained on 2 different frequencies; either 1 session every second week or 1 session per week in addition to normal off-season activity. Their aerobic fitness was evaluated before and after the training intervention using 2 protocols; a 20-m shuttle run test and a graded $\dot{V}O_2\text{max}$ test by running on a treadmill. To investigate whether 1 session every second week or 1 session per week is sufficient for maintaining aerobic fitness, changes in aerobic fitness after the training intervention was compared between the 2 training regimes.

Subjects

Eighteen male semiprofessional soccer players (age, 18–26 years) at the second and third highest division in Norway volunteered to participate in this study. After the initial 20-m shuttle run test and $\dot{V}O_2\text{max}$ test, the subjects were randomized into 1 HIT session every second week (HIT 0.5) or 1 HIT session per week (HIT 1). One subject did not complete the training intervention due to illness and was excluded from the study. Eight subjects from the HIT 0.5 group and 9 subjects from the HIT 1 group completed the study. Baseline characteristics of the HIT 0.5 group (21.8 ± 2.2 years; age range, 18–24 years; height, 179 ± 4 cm; and body weight, 74.9 ± 6.2 kg) were not different from the HIT 1 group (19.7 ± 3.1 years, age range 18–26 years; height, 183 ± 8 cm; and body weight, 74.1 ± 7.6 kg). During the last 3 months before the study took place, the subjects normally trained 4 organized training sessions per week in addition to 1 soccer match every week. The project was approved by the local ethical committee of Lillehammer University College and written informed consent was gained from all subjects.

Procedures

To reduce the impact of nocturnal changes on performance, testing of the individual subject was standardized to approximately the same time of day. The players were instructed to refrain from intense exercise the day preceding testing and to consume the same type of meal before each test. They were not allowed to eat during the hour preceding the test or to consume coffee or other products containing caffeine during the preceding 3 hours. All subjects performed a 10-minute individual warm-up followed by a 20-m shuttle run test. The test protocol and audio CD were produced by Loughborough University and National Coaching Foundation (2002). The test protocol was followed according to the manufacturer's instructions. In short, the shuttle runs were

between 2 markers placed 20 m apart on a wooden floor of an indoor sports arena. One foot had to cross the marker when a beep signal was played from the CD player, and the pace increased progressively as the frequency of the beep signal increased. If the subjects failed to reach the marker before the beep signal was played on 2 consecutive trails, the subject was withdrawn from the test. $\dot{V}O_2\text{max}$ was estimated from the shuttle run performance as described by Ramsbottom et al. (18). Two to 5 days after the 20-m shuttle run test, the subjects performed a $\dot{V}O_2\text{max}$ test on a motorized treadmill (Woodway; Wankesha, Wisconsin, USA). The warm-up consisted of 10-minute easy running on 5.5% incline followed by 3 bouts with stepwise increased speed (2 bouts; $1 \text{ km} \cdot \text{h}^{-1}$ every 10 seconds and 1 bout; $1 \text{ km} \cdot \text{h}^{-1}$ every 5 seconds from 10 to $17 \text{ km} \cdot \text{h}^{-1}$). After the warm-up, treadmill incline was adjusted to 5.3%, initial speed was $10 \text{ km} \cdot \text{h}^{-1}$, and speed was increased by $1 \text{ km} \cdot \text{h}^{-1}$ every minute until exhaustion. Pulmonary gas exchange was measured continuously (Oxycon Pro; Eric Jaeger, Hoechst, Germany) using mixing chamber, and values were averaged every 30 seconds. Venous blood was sampled from the

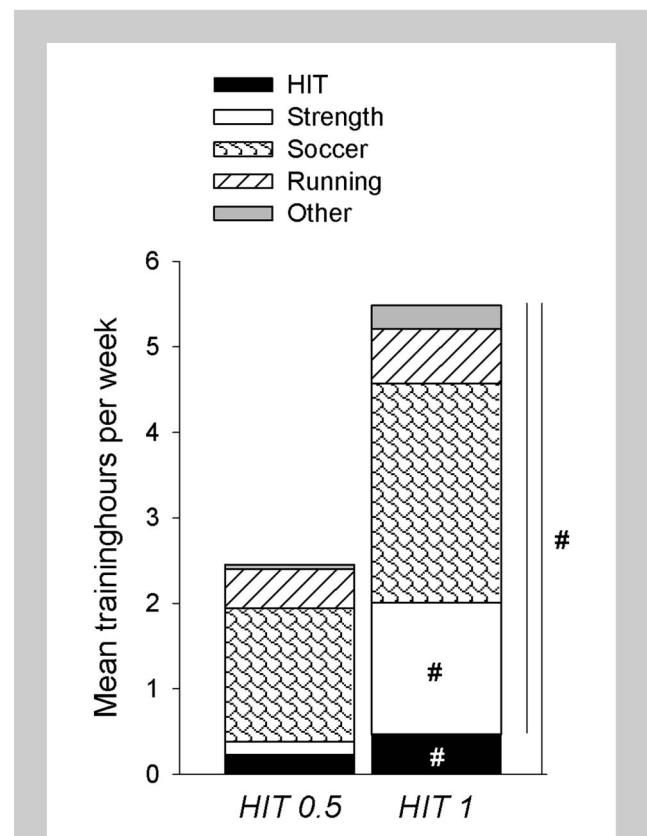


Figure 1. Mean training hours per week divided in 5 categories: The high-intensity interval training (HIT), which was the training intervention and strength, soccer, running, and other that were voluntary additional workouts were logged by the subjects. HIT 0.5 is the group that trained 1 HIT session every second week and HIT 1 is the group that trained 1 HIT session per week. # $p < 0.01$ is different than HIT 0.5.

fingertip at exhaustion and analyzed for plasma lactate concentration (Lactate Pro LT-1710; Arcaray Inc., Kyoto, Japan). At exhaustion, none of the subjects had a respiratory exchange ratio (RER) below 1.0 or blood lactate concentration below $8 \text{ mmol} \cdot \text{L}^{-1}$ indicating that all subjects ran until exhaustion. Maximum treadmill velocity ($V_{\max}$) was calculated as second last velocity along with the increase in velocity ($1 \text{ km} \cdot \text{h}^{-1}$) multiplied by time fraction spent on last velocity. $\dot{V}O_{2\max}$ was calculated as the average of the 2 highest $\dot{V}O_2$ measurements. Rate of perceived exertion was measured by Borg's 6-20 scale (2).

The HIT sessions consisted of 5 bouts of 4 minutes of high-intensity (87–97% of peak heart rate) treadmill running. Similar protocol has been shown to improve $\dot{V}O_{2\max}$ more than lower intensity training, and the protocol has frequently been used on soccer players (8,9). Heart rate was monitored using radio telemetry (Polar; Pro Electro Oy, Kempele, Finland). All training sessions were supervised and logged by a skilled instructor. All subjects were free to perform additional workouts, which they logged and categorized in strength, soccer, running, and other exercise training. Mean additional workouts were 2.2 ± 1.3 and 5.0 ± 2.0 hours per week for HIT 0.5 and HIT 1 groups, respectively with strength training as the main contributor to the difference between the groups ($p < 0.01$, Figure 1).

Statistical Analyses

Data are expressed as mean $\pm$ SD. Dependent variable in this study was HIT frequency (HIT 0.5 or HIT 1), and primary independent variables were $V_{\max}$, $\dot{V}O_{2\max}$, and

distance covered during the 20-m shuttle run test. Independent sample t -test was used when comparing baseline values of independent variables between HIT 0.5 and HIT 1. To analyze the effect of HIT 0.5 and HIT 1 on performance during the $\dot{V}O_{2\max}$ test and the 20-m shuttle run test, a paired sample t -test comparing values after the training intervention with before the training intervention was used. Effect size (ES) was calculated as Cohen's d to compare the practical significance of change in $\dot{V}O_{2\max}$ and distance covered during 20-m shuttle run test among the 2 groups. The criteria to interpret the magnitude of the ES were the following: 0.0–0.2 trivial, 0.21–0.6 small, 0.61–1.2 moderate, 1.21–2.0 large, and >2.0 very large (11). Pearson's product correlation was used when evaluating the relation between $\dot{V}O_{2\max}$ measured during the $\dot{V}O_{2\max}$ test and $\dot{V}O_{2\max}$ estimated from the 20-m shuttle run test. Effect size of correlation coefficients were defined as $r < 0.1$, trivial; 0.1–0.3, small; 0.3–0.5, moderate; 0.5–0.7, large; 0.7–0.9, very large; 0.9 nearly perfect; and 1.0 perfect (11). Bland-Altman plot was used for assessing agreement between the methods. A p value ≤ 0.05 was considered significant. SPSS 20 for Windows software (IBM Corp, Armonk, NY, USA) was used for statistical analysis.

RESULTS

Before the training period, $V_{\max}$, $\dot{V}O_{2\max}$, RER, Borg score, and peak heart rate from the $\dot{V}O_{2\max}$ test were not different between the 2 groups. Blood lactate, however, was 17% higher in the HIT 1 group than the HIT 0.5 group ($p = 0.02$, Table 1). The distance covered during the 20-m shuttle run test and heart rate at exhaustion was not different between the 2 groups (Table 1).

After the training period, $V_{\max}$, $\dot{V}O_{2\max}$, and RER from the $\dot{V}O_{2\max}$ test were not different from before the training period in either of the groups (Table 1 and Figure 2A). Borg score, however, was 0.7 points lower after the training period for the HIT 0.5 group ($p = 0.05$, Table 1). The distance covered during the 20-m shuttle run test was not different from before the training period for the HIT 0.5 group but was $8 \pm 6\%$ shorter after the training period for the HIT 1 group ($p \leq 0.05$, Table 1 and Figure 1B) and $6 \pm 8\%$ shorter when the groups were pooled ($p \leq 0.05$). Mean ES of the relative changes in $\dot{V}O_{2\max}$ and distance covered during the

TABLE 1. Results (mean $\pm$ SD) from $\dot{V}O_{2\max}$ and 20-m shuttle run tests before and after the training intervention for the group performing HIT 1 session every second week (HIT 0.5) and 1 session per week (HIT 1).*

	HIT 0.5 (n = 8)		HIT 1 (n = 9)	
	Before	After	Before	After
$\dot{V}O_{2\max}$ test				
$V_{\max}$ ($\text{km} \cdot \text{h}^{-1}$)	14.4 ± 1.0	14.6 ± 0.9	14.7 ± 0.6	14.7 ± 0.9
$\dot{V}O_{2\max}$ ($\text{ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$)	63.4 ± 5.9	64.0 ± 5.9	65.6 ± 2.1	64.3 ± 1.3
RER	1.14 ± 0.04	1.11 ± 0.05	1.12 ± 0.04	1.11 ± 0.04
Lactate ($\text{mmol} \cdot \text{L}^{-1}$)	$12.0 \pm 2.2^\dagger$	10.2 ± 1.5	$14.5 \pm 1.9^\dagger$	13.4 ± 2.8
Borg score	$18.8 \pm 0.9^\ddagger$	18.1 ± 0.8	19.3 ± 1.0	18.4 ± 1.5
HR_{peak} ($\text{b} \cdot \text{min}^{-1}$)	203 ± 12	200 ± 8	200 ± 6	196 ± 8
20-m shuttle run test				
Distance (m)	2335 ± 390	2213 ± 345	$2531 \pm 106^\ddagger$	2327 ± 96
$\dot{V}O_{2\max}$ ($\text{ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$)	57.1 ± 5.1	54.9 ± 4.8	$59.3 \pm 1.5^\ddagger$	56.4 ± 1.3
HR ($\text{b} \cdot \text{min}^{-1}$)	199 ± 6	201 ± 5	$194 \pm 8^\ddagger$	198 ± 7

*HIT = high-intensity interval training; $\dot{V}O_{2\max}$ = maximal oxygen uptake; $V_{\max}$ = maximal treadmill velocity; RER = respiratory exchange ratio; HR = heart rate.

$^\dagger p \leq 0.05$ between groups before the training period.

$^\ddagger p \leq 0.05$ after vs. before the training period.

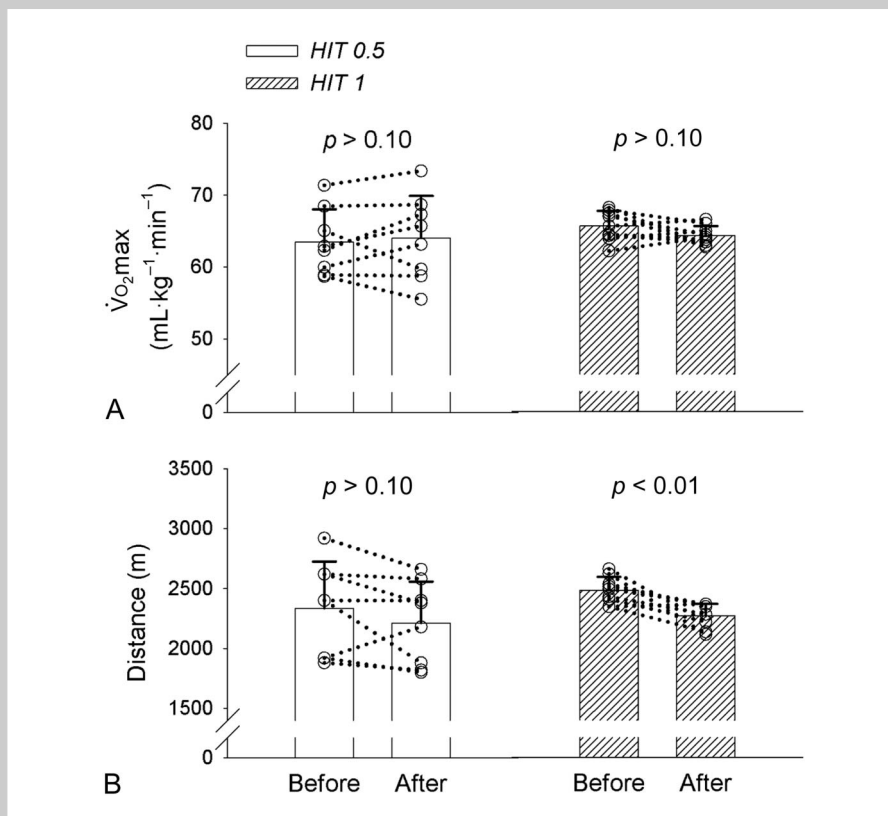


Figure 2. A) Mean (bars) and individual (circles) values of $\dot{V}O_{2\max}$ before and after the training period for those training high-intensity interval training (HIT) 1 session every second week (HIT 0.5) and 1 session per week (HIT 1). B) Mean (bars) and individual (circles) values of distance covered during the 20-m shuttle run test before and after the training period. Difference between before and after the training period were indicated by p values.

20-m shuttle run test revealed a small effect of HIT 0.5 vs. HIT 1 ($ES \leq 0.6$).

$\dot{V}O_{2\max}$ estimated from the 20-m shuttle run test correlated with the $\dot{V}O_{2\max}$ measured during the $\dot{V}O_{2\max}$ test ($r = 0.85$, $p < 0.01$) but underestimated $\dot{V}O_{2\max}$ with $10 \pm 3\%$ (Figure 3A). A Bland-Altman plot showed little

declines more during deconditioning than in trained subjects with a lower initial $\dot{V}O_{2\max}$ (6), it would be plausible that the soccer players in this study would decline in $\dot{V}O_{2\max}$ as a result of reduced exercise training during off-season. However, $\dot{V}O_{2\max}$ did not decline in any of the groups indicating that HIT once every second week when added to normal off-

season activity was sufficient for maintaining maximal aerobic capacity. Considered the players' normal off-season activity, mean training hours per week of soccer specific training and running was not different between the groups during the training intervention. The higher quantity of total training hours for the HIT 1-group was due to higher quantity of strength training and HIT sessions (Figure 1). Although concurrent strength and endurance training may improve performance in short-duration

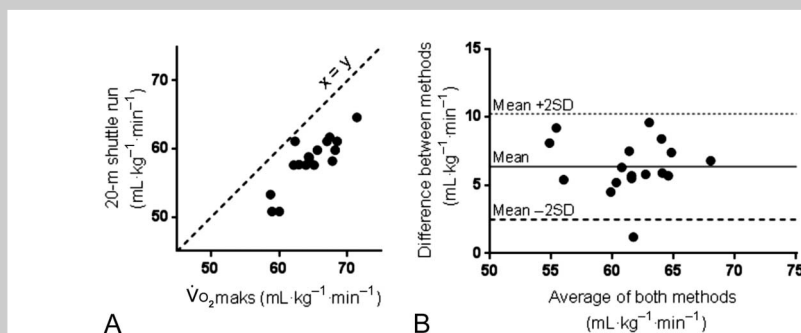


Figure 3. A) Correlation between $\dot{V}O_{2\max}$ from the $\dot{V}O_{2\max}$ test was estimated from the 20-m shuttle run test. B) Bland-Altman plot showing difference vs. average of values from the $\dot{V}O_{2\max}$ test and the 20-m shuttle run test with 95% limits of agreement.

exercise in highly trained athletes (19), numerous studies report no effect of strength training on $\dot{V}O_2\text{max}$ development (20,21). Thus, the larger amount of strength training in HIT 1-group is not likely to affect the $\dot{V}O_2\text{max}$ data. The subjects did not register training intensity for the self-reported additional training, only training hours. Although unlikely, we cannot exclude that any difference in intensity for the additional training may have masked possible differences in training response between the groups or that the additional training was sufficient alone for maintaining $\dot{V}O_2\text{max}$.

The minimum training intensity that enhances $\dot{V}O_2\text{max}$ is highly dependent on initial $\dot{V}O_2\text{max}$ (25), and optimal exercise intensity for trained athletes is probably close to $\dot{V}O_2\text{max}$ (16). The importance of training intensity in maintaining $\dot{V}O_2\text{max}$ is further emphasized by Houmard et al. (12) who found that a 70% reduction in training volume in endurance runners did not result in reduced $\dot{V}O_2\text{max}$ when the training was performed at an intensity between 75 and 95% of $\dot{V}O_2\text{max}$. Evaluated from the heart rate and treadmill velocity during the HIT compared with the $\dot{V}O_2\text{max}$ test in this study, intensity during HIT was probably close to $\dot{V}O_2\text{max}$ for both groups. This study therefore suggests that 1 session of HIT every second week in addition to off-season activity is sufficient for maintaining $\dot{V}O_2\text{max}$ as long as training intensity is kept close to $\dot{V}O_2\text{max}$.

Although $\dot{V}O_2\text{max}$ is a decisive factor for the distance soccer players covers during matches (8), the $\dot{V}O_2\text{max}$ test is often performed in laboratories with linear running on a treadmill. This test is therefore less soccer-specific than the shuttle run test, which involves repeated accelerations, decelerations, and has been shown to be related to specific soccer movements (1,5,7,17). In theory, the larger strength- and soccer training volume in the HIT 1-group may induce superior performance in the shuttle run test. It has been showed that soccer players increases their sprint and jump performance after a strength training period, which was accompanied by increased maximal strength (22,23). Improved strength, sprint, and jump abilities can, in theory, enhance the shuttle run performance through increased efficiency in the changes of running direction and postponed fatigue caused by the multiple decelerations and accelerations. However, this was not the case and indeed, it was only the HIT 1 group that had a significant reduction in shuttle run performance. That being said, there was a great interindividual variation in the HIT 0.5 group with 1 player who increased his 20-m shuttle run distance with 14%, and the ES showed only a small difference between the 2 training regimes on shuttle run performance. Furthermore, when groups were pooled, there was a significant 6% reduction in shuttle run performance after the intervention period. This indicates that difference in strength training between the groups did not have effect on shuttle run performance. For semiprofessional soccer players, the amount of soccer-specific training is reduced during off-season. The combination of sufficient endurance exercise to maintain aerobic capacity but a reduced soccer-specific training might be

the explanation of maintained $\dot{V}O_2\text{max}$ but reduced performance in the 20-m shuttle run test during the off-season in this study.

The estimated $\dot{V}O_2\text{max}$ from the 20-m shuttle run had a very large correlation with $\dot{V}O_2\text{max}$ measured during the graded treadmill $\dot{V}O_2\text{max}$ test. This agrees with previous findings that values of $\dot{V}O_2\text{max}$ -estimated 20-m shuttle run is related to $\dot{V}O_2\text{max}$ measured during a graded $\dot{V}O_2\text{max}$ test (14,24). However, product moment correlation coefficient is not an indicator of agreement between the 2 methods, and the Bland-Altman method is more appropriate. Our results showed that the 20-m shuttle run test underestimated $\dot{V}O_2\text{max}$ by approximately 10% compared with the direct method, and that there were no agreement between the 2 methods.

There are some limitations in this study. First, although all training hours outside the organized HIT training was logged, the training intensity of the self-inflicted training was not logged and could therefore be a confounding factor when comparing the training effect between the groups. Another limitation is that we did not have any control group not participating in organized HIT training. Therefore, we cannot make conclusions about the effect of additional HIT compared with no additional HIT training. We can only make conclusions based on 2 different frequencies of HIT training.

In conclusion, both 1 session of HIT per week and 1 session of HIT every second week in addition to normal off-season activity maintained $\dot{V}O_2\text{max}$. However, performance in 20-m shuttle run, which is a more soccer-specific fitness test than $\dot{V}O_2\text{max}$ test, was slightly reduced when both groups were pooled. This study therefore suggests that more than 1 HIT session per week is needed for maintaining aerobic fitness among semiprofessional soccer players during the off-season period.

PRACTICAL APPLICATIONS

For semiprofessional soccer players, the off-season of 4–7 weeks represents a period with less exercise training and potentially reduced aerobic capacity. Because these players usually have more obligations outside the soccer field than professional players, cutting training hours to a minimum during this period is desired. HIT is a potent method for maintaining aerobic capacity using less training hours. In this study, we have compared 2 different HIT frequencies. This study suggests that adding 1 HIT session every second week to normal off-season activity is sufficient for maintaining $\dot{V}O_2\text{max}$. However, it seems that this training frequency or even when it is doubled to 1 session per week is not enough for maintaining performance during shuttle run tests, which is regarded as a more soccer-specific than the $\dot{V}O_2\text{max}$ test. This knowledge is also useful in training periods where the coach wants to emphasis development of other qualities (e.g., strength/power abilities or tactical issues) than $\dot{V}O_2\text{max}$ and therefore will spend as little time as possible on maintaining $\dot{V}O_2\text{max}$. In the latter scenario, 1 HIT session every second week for some weeks seems to maintain $\dot{V}O_2\text{max}$. HIT is

a time-efficient training mode that can be recommended for soccer players between 2 competition seasons.

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